force per unit volume

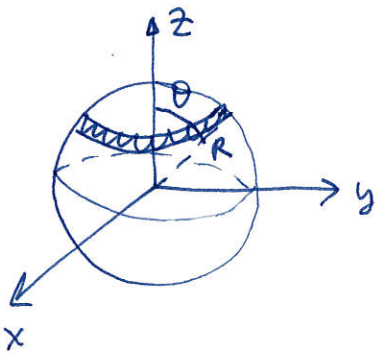
Example

$$\vec{f} = \vec{\nabla}\cdot\overleftrightarrow{\sigma} - \varepsilon_0\mu_0 \frac{\partial \vec{S}}{\partial t}$$

total $\vec{F}$ on volume V

$$\vec{F} = \int_V \vec{\nabla}\cdot\overleftrightarrow{\sigma}\, dV - \varepsilon_0\mu_0 \int \frac{\partial \vec{S}}{\partial t}\, dV = \oint_{\substack{\text{Area}\\ \text{enclosing } V}} \overleftrightarrow{\sigma}\cdot d\vec{a} - \varepsilon_0\mu_0 \underbrace{\frac{\partial}{\partial t}\int \vec{S}\cdot dV}_{\substack{\emptyset \\ \text{for static}\\ \text{fields}}}$$

$\sigma_{\alpha\beta}$ = force per unit area on the α-th direction acting on element of surface oriented in β-th direction.



Uniformly charged sphere (R, Q).
Determine force on northern hemisphere

1) Hemisphere $d\vec{a} = R\,d\theta \cdot R\sin\theta\, d\varphi\ \hat{r}$

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{r}$$

$$\hat{r} = \sin\theta\cos\varphi\ \hat{x} + \sin\theta\sin\varphi\ \hat{y} + \cos\theta\ \hat{z}$$

$$\sigma_{\alpha\beta} = \varepsilon_0 E_\alpha E_\beta + \frac{1}{\mu_0} B_\alpha B_\beta - \frac{1}{2}\delta_{\alpha\beta}\left(\varepsilon_0 \vec{E}^2 + \frac{1}{\mu_0}\vec{B}^2\right)$$

$$\sigma_{xx} = \varepsilon_0\left[E_x^2 - \frac{1}{2}\left(E_x^2+E_y^2+E_z^2\right)\right] + \frac{1}{\mu_0}\left[B_x^2 - \frac{1}{2}\left(B_x^2+B_y^2+B_z^2\right)\right] =$$

$$= \frac{\varepsilon_0}{2}\left(E_x^2 - E_y^2 - E_z^2\right) + \frac{1}{2\mu_0}\left(B_x^2 - B_y^2 - B_z^2\right)$$

$$\sigma_{xy} = \varepsilon_0 E_x E_y + \frac{1}{\mu_0} B_x B_y \ \ldots$$

Clearly the net force $\vec{F} = F_z \hat{z}$

$$F_z = \int \sigma_{zx}\, da_x + \sigma_{zy}\, da_y + \sigma_{zz}\, da_z$$

$$\sigma_{zx} = \varepsilon_0 E_z E_x = \varepsilon_0 \left(\frac{Q}{4\pi\varepsilon_0 R^2}\right)^2 \sin\theta \cos\theta \cos\varphi$$

$$\sigma_{zy} = \varepsilon_0 E_z E_y = \varepsilon_0 \left(\frac{Q}{4\pi\varepsilon_0 R^2}\right)^2 \sin\theta \cos\theta \sin\varphi$$

$$\sigma_{zz} = \frac{\varepsilon_0}{2}\left(E_z^2 - E_x^2 - E_y^2\right) = \frac{\varepsilon_0}{2}\left(\frac{Q}{4\pi\varepsilon_0 R^2}\right)^2 \left(\cos^2\theta - \sin^2\theta\right)$$

$$da_x = R^2 \sin\theta\, d\theta\, d\varphi\, (\sin\theta \cos\varphi)$$

$$da_y = R^2 \sin\theta\, d\theta\, d\varphi\, (\sin\theta \sin\varphi)$$

$$da_z = R^2 \sin\theta\, d\theta\, d\varphi \cos\theta$$

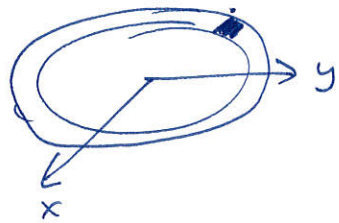
$$F_z^{(1)} = \varepsilon_0 R^2 \left(\frac{Q}{4\pi\varepsilon_0 R^2}\right)^2 \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin\theta \left\{ \frac{\cos^2\theta - \sin^2\theta}{2} \cos\theta + \right.$$

$$\left. + \sin\theta \cos\theta \underline{\sin\varphi}\, \sin\theta \underline{\sin\varphi} + \sin\theta \cos\theta \underline{\cos\varphi}\, \sin\theta \underline{\cos\varphi} \right\} =$$

$$= \varepsilon_0 R^2 \left(\frac{Q}{4\pi\varepsilon_0 R^2}\right)^2 (2\pi) \int d\theta \sin\theta \underbrace{\left(\sin^2\theta \cos\theta + \tfrac{1}{2}(\cos^2\theta - \sin^2\theta)\cos\theta\right)}_{\frac{1}{2}(\cos^2\theta + \sin^2\theta)\cos\theta}$$

$$= \pi \varepsilon_0 \left(\frac{Q}{4\pi\varepsilon_0 R}\right)^2 \underbrace{\int_0^{\pi/2} d\theta \sin\theta \cos\theta}_{\int_0^{\pi/2} d(\sin\theta) \sin\theta = \frac{1}{2}} = \frac{\pi\varepsilon_0}{2} \frac{Q^2}{16\pi^2\varepsilon_0^2 R^2} = \frac{Q^2}{4\pi\varepsilon_0 \cdot 8R^2}$$

2) Base $\left(\theta = \frac{\pi}{2}\right)$ $\quad d\vec{a} = r\,dr\,d\varphi\,(-\hat{z}) = da_z\,\hat{z}$



$$\vec{E}\cdot 4\pi r^2 = \frac{1}{\varepsilon_0}\rho\,\frac{4\pi}{3}r^3$$

$$E(r) = \frac{\rho r}{3\varepsilon_0} = \frac{Q}{\frac{4\pi}{3}R^3}\,\frac{r}{3\varepsilon_0} = \frac{Qr}{4\pi\varepsilon_0 R^3}$$

$$\vec{E}(\vec{r}) = \frac{Q}{4\pi\varepsilon_0 R^3}\,\vec{r} = \frac{Qr}{4\pi\varepsilon_0 R^3}\sin\theta\,(\cos\varphi\,\hat{x} + \sin\varphi\,\hat{y})\Big|_{\theta=\frac{\pi}{2}}$$

(Note $E_z = 0$ in equatorial plane.)

$\Rightarrow$ $\sigma_{zx} = \sigma_{zy} = 0$

$$\sigma_{zz} = \frac{\varepsilon_0}{2}\left(\underset{\overset{\shortparallel}{0}}{E_z^2} - E_x^2 - E_y^2\right) = -\frac{\varepsilon_0}{2}\left(\frac{Qr}{4\pi\varepsilon_0 R^3}\right)^2$$

$$F_z^{(2)} = +\frac{\varepsilon_0}{2}\left(\frac{Q}{4\pi\varepsilon_0 R^3}\right)^2 \int_0^{2\pi} d\varphi \int_0^R dr\, r\,(r^2) =$$

$$= \pi\varepsilon_0\left(\frac{Q}{4\pi\varepsilon_0 R^3}\right)^2 \frac{1}{4}R^4 = \frac{Q^2}{4\pi\varepsilon_0\cdot 16R^2}$$

$$F_z^{\text{total}} = \frac{1}{4\pi\varepsilon_0}\,\frac{Q^2}{8R^2}\left(1+\frac{1}{2}\right) = \frac{1}{4\pi\varepsilon_0}\,\frac{3Q^2}{16R^2}$$

(3) Let us choose Area to be equatorial plane with $r\in(0,\infty)$ + hemisphere at $R=\infty$ (so that $\vec{E}=0$ there).

$\Rightarrow F^{tot} = F^{(2)} + F^{(3)} = F^{(1)} + F^{(2)}$ same thing!

$$F^{(3)} = \int_{r=R}^{\infty} \sigma_{zz}\,da_z = -\frac{\varepsilon_0}{2}\int_R^{\infty}(-r\,dr\,d\varphi)\left(\frac{Q}{4\pi\varepsilon_0 r^2}\right)^2 =$$

$$= \frac{2\pi Q^2}{16\pi^2\varepsilon_0}\int_R^{\infty}\frac{dr\,r}{r^4} = \frac{Q^2}{16\pi\varepsilon_0}\left(-\frac{1}{2r^2}\right)\Big|_R^{\infty} = \frac{Q^2}{4\pi\varepsilon_0\cdot 8R^2} = F^{(1)}$$